Supplementary Material

Histological, chemical and gene expression differences between western redcedar seedlings resistant and susceptible to cedar leaf blight

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1 Supplementary Methods

1.1 Confirmation of *D. thujina* infection during the screening of *T. plicata* seedlings

Samples used for scanning electron microscopy (SEM) analyses were collected from four ~3 mm leaf sections (two from the uppermost branch, one from the middlemost branch and one from the lowermost branch) of five seedlings per family on the fourth day of field deployment. The samples were fixed overnight in 2.5% glutaraldehyde (prepared in Sörensen buffer pH 7.2; Ruzin 1999). The following day, the fixative was washed out of the samples by two consecutive changes of 30 min each in Sörensen buffer. The samples were then dehydrated by 30 min changes in an ascending ethanol series (50%, 70%, 80%, 90% and twice in 100%). Next, samples were critical point dried in a Leica EM CPD300 system (Leica Microsystems Inc., Richmond Hill ON, Canada). The dried samples were then gold coated in an Edwards S150B Sputter Coater (Edwards Canada, Quebec QC, Canada), and photographed at 1000X in a Hitachi S-3500N Scanning Electron Microscope (Hitachi High-Technologies Canada Inc., Toronto ON, Canada). The length and width of the photographed spores were measured using ImageJ v. 1.46 (Schneider et al., 2012). Those measurements as well as the general morphology of the spores were compared to the published description of *D. thujina* (Durand, 1913, Kope, 2000).

The anatomy of the foliage with symptoms was studied using the modified clearing and staining technique of McLean and Byth (1981) used by Liberato et al. (2005). In brief, leaf samples of the same size as those used for SEM, but this time with lesions present, were cleared for 24 h in a 1:1 v/v mix of anhydrous ethanol and 100% acetic acid at room temperature (RT). The cleared samples were then washed with three 5-min changes in distilled water. Free-hand sections were cut and mounted on microscope slides with 0.1% w/v aniline blue prepared in 85% lactic acid. The sections were examined and photographed at 63X and 400X magnification using a Zeiss microscope fitted with a RTKE SPOT camera (SPOT Imaging, Sterling Heights MI, USA) and SPOT v. 4.6 software.

1.2 Histological characterization of uninfected *T. plicata* seedlings

All samples collected for histological analyses, except those for stomatal density measurements, were fixed as described in section 1.1. The following day, the fixative was washed out of the samples with two changes of 20 min each in Sörensen buffer. The samples were then dehydrated in the same ethanol series used for the SEM preparation, but changing solutions every 20 min. After the last 100% ethanol step, the samples were changed to a 1:1 v/v ethanol-*n*BGE (*n*-butyl glycidyl ether) solution for 30 min, then to 100% *n*BGE for 30 min, next to a 1:1 v/v *n*BGE-Quetol 651 for 2 h, and lastly to 100% Quetol

651 overnight. The next day, the samples were polymerized in fresh 100% Quetol 651 (Takagi and Sato, 1979) for 24 h at 60℃. Embedded samples were sectioned at 5 µm using a Sorvall JB-4 Microtome and fixed to microscope slides on a hot plate for further processing. Two sets of sections per sample were produced, one for general histology staining with Toluidine Blue O, and another to quantify lignification of epidermal cell walls stained with phloroglucinol.

Samples for general histology were covered for 5 min with 1.0 % w/v Toluidine Blue O (dissolved in 1% aqueous sodium borate), rinsed with distilled water, and then washed for 50 s in 70% ethanol and for 1 min in 100% ethanol. The stained samples were dried at RT and mounted using Permount™ (Fisher Chemical™, Ottawa ON, Canada). Sections were photographed at 100X and 1000X magnifications using the same microscope and with the same camera mentioned in section 1.1. Digitized images were measured with the same software in section 1.1.

For staining lignification of epidermal cell walls, microscope slides with sections were flooded for 5 min with 1% w/v phloroglucinol (prepared in 70% ethanol). Excess phloroglucinol was removed and the slides covered for 1 min with hydrochloric acid (HCl). The excess HCl was removed, and the sections dried at RT. The samples were then mounted as before and photographed within 3 h as described above.

Stomatal densities were studied on leaf impressions produced using transparent nail polish. Two coats were applied to the area of interest and dried for 30 min at RT. The impressions were then removed with transparent tape, mounted on microscope slides and covered with coverslips. The impressions were next photographed at 100X magnification.

Individual continuous histological variables were analyzed according to the following split-plot statistical model:

$$Y_{ijklm} = \mu + \alpha_i + \beta_{j(i)} + \gamma_{k(j)} + \delta_l + (\alpha\delta)_{il} + (\beta\delta)_{jl} + (\gamma\delta)_{kl} + \varepsilon_{ijklm}$$

Where:

μ	=	Mean value of the variable of interest.
α_i	=	Effect of the i^{th} resistance class.
$\beta j(i)$	=	Effect of the j^{th} family within the i^{th} resistance class.
$\gamma k(j)$	=	Main plot effect (k^{th} seedling within j^{th} family).
δ_l	=	Effect of the l^{th} branch position per seedling.
$(\alpha\delta)_{il}$	=	Effect of the interaction of the l^{th} branch position in the i^{th} resistance class.
$(\beta\delta)_{jl}$	=	Effect of the interaction of the l^{th} branch position in the j^{th} family.
$(\gamma\delta)_{kl}$	=	Effect of the interaction of the k^{th} seedling in the l^{th} branch position.
ε_{ijklm}	=	Residual error effect.

Model parameters and mean squares were calculated in R (R Core Team, 2015).

1.3 Chemical composition analyses

Samples for carbon and nitrogen analyses were dried at 60℃ for 48 h, and then ground with a Wig-L-Bug grinding mill (International Crystal Laboratories, NJ USA). The samples were then dried overnight before being packaged. The packaging was done in tin pans (10×10mm; Elemental

Microanalysis Ltd., Devon UK) and the measurements were carried out in a FlashEA® 1112 Nitrogen and Carbon Analyzer (Thermo Scientific™ Wilmington DE, USA).

Terpenes were quantified from foliage samples ground in liquid nitrogen with a chilled pestle and mortar. Four millilitres of methanol were mixed with 0.25-0.50 g of ground material for 48 h to extract the terpenes. *T. plicata* extractive standards were prepared mixing 0.05-0.50 g of the respective standard with 100 mL of methanol. The amount per standard varied depending on the anticipated concentrations in the samples analyzed. Samples and standards were analyzed using 30 m × 0.25 mm, d_f 0.25 µm capillary GC columns in a Clarus Gas Chromatograph (PerkinElmer Inc., Waltham MA, USA) following the manufacturer's instructions.

Mineral analyses were completed by mixing 0.25 g of fine milled foliage with 4.0 mL digestion reagent (150 µg scandium·L^{-1} prepared in 70% nitric acid) in 15 mL quartz tubes. The digestion was carried out in an Ultrawave Microwave Digestion System (Milestone Inc., Shelton CT, USA) following the manufacturer instructions, and elements were measured with inductively coupled plasma optical emission spectroscopy analysis using the scandium from the digestion step as standard.

Statistical analyses of variables retained by the changepoint analyses

Individual chemical variables retained by the changepoint analyses of the categorical stability selection outputs were analyzed according to the following statistical fixed-effects factorial model:

$$Y_{ijklm} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

Where:

μ	=	Mean value of the variable of interest.
α_i	=	Effect of the i^{th} family.
β_j	=	Effect of the j^{th} infection treatment.
$(\alpha\beta)_{ij}$	=	Effect of the interaction of the i^{th} family in the j^{th} infection treatment.
ε_{ijk}	=	Residual error effect.

Model parameters and mean squares were calculated in R (R Core Team, 2015).

1.4 Gene expression analyses

RNA extraction

The following, modified version, of the protocol by Rajakani et al. (2013) was used for *RNA* extraction. One-and-a-half grams of the sampled foliage were ground in a pre-chilled mortar and pestle using liquid nitrogen. The ground material was transferred to a 50 mL Nalgene™ Oak Ridge high-speed centrifuge tube (Life Technologies Inc., Burlington ON, Canada) and the mortar rinsed with liquid nitrogen. Ground foliage with liquid nitrogen was then placed in a -20°C freezer to let the nitrogen evaporate. Once the material was dry, 20 mL of preheated 2% CTAB with activated charcoal were added to each sample (200 mM Tris-Cl pH 8.0, 50 mM EDTA pH 8.0 and 2.5 M NaCl, 0.05% activated charcoal, 1.5% PVPP, 3% β-mercaptoethanol; the last three components were added just before using the extraction buffer). The samples were thoroughly vortexed and incubated at 65°C for 15 min with intermittent shaking. The mixture was then cooled to RT and centrifuged at 17,211 × *g* for 20 min at 4°C. The supernatant was recovered in a new 50 mL tube and combined with an equal volume of chloroform, mixed by inversion and centrifuged at 17,211 × *g* for 20 min at 4°C. The previous purification step was repeated once more (for a total of two chloroform extractions). The supernatant from the last extraction was then transferred to a new 50 mL tube, mixed with 1/4 volume of 10M lithium chloride and incubated overnight at 4°C.

The next day, samples were centrifuged at 20,199 × *g* for 30 min at 4°C. The supernatant was discarded while the pellet was re-suspended in a mixture of 500 µL SSTE (1M NaCl, 0.5% SDS, 10mM Tris HCl pH 8.0, 1mM EDTA pH 8.0) and 400 µL phenol (pH 8.0). The suspension was transferred to a 1.5 mL Eppendorf® tube to which 100 µL of chloroform were added. The solution was vortexed and centrifuged at 18,407 × *g* for five min at RT. The supernatant was transferred to a new 1.5 mL tube, mixed with 300 µL of chloroform, vortexed and centrifuged again for 5 min at 18,407 × *g* at RT. The supernatant from the last extraction was placed in a new 1.5 mL tube to which 1/10 volume of 3M sodium acetate and an equal volume of 100% isopropanol were added. The solution was incubated for 25 min at -20°C, and then centrifuged at 20,626 × *g* at 4°C for 33 min. The supernatant was discarded, 1.0 mL of 75% chilled ethanol was added to the pellet and mixed by inversion. The mixture was centrifuged for 5 min at 20,626 × *g* at 4°C, the supernatant was discarded once more to ensure that no ethanol remained (so that the pellet dried at RT). The total extracted *RNA* was suspended in 50 µL of DEPC-treated water, quantified in a NanoDrop™ 2000 Spectrophotometer (Thermo Scientific™ Wilmington DE, USA), and checked for integrity in a 1X MOPs, 1.0% formaldehyde-agarose gel.

Trimming of the FASTQ files and annotation of the reference transcriptome

The pipeline used for the bioinformatics analyses is shown in Supplementary Figure 1 below. Trimming of the FASTQ Illumina® files was completed in Trimmomatic v. 0.33 (Bolger et al., 2014) with the following settings: ILLUMINACLIP: TruSeq2-PE-v2.fa:2:30:10 HEADCROP:14 SLIDINGWINDOW:4:15 MINLEN:36 TOPHRED33. The TruSeq2-PE-v2.fa was a custom-made version of the TruSeq2-PE.fa of Trimmomatic produced by adding over-represented sequences reported by FastQC.

The annotation carried out in Trinotate v. 2.0.2 (http://trinotate.github.io) consisted of: 1) prediction of coding regions with TransDecoder v. 2.0.1. (http://transdecoder.github.io), 2) search for annotations for those predicted coding regions (including Gene Ontology term enrichment, Ashburner et al. 2000, The Gene Ontology Consortium 2015; and EggNOG annotation, Huerta-Cepas et al. 2016) in the Swiss-Prot/TrEMBL and UniRef90 (The UniProt Consortium, 2017) companion databases of Trinotate using BLAST+ v. 2.2.28 (Camacho et al., 2009), 3) search for protein domains in Pfam (Finn et al., 2016) using HMMER v. 3.1b1 (http://hmmer.org), and 4) search for signal peptides with SignalP v. 4.0 (Petersen et al., 2011), transmembrane regions with TMHMM v. 2.0 (http://www.cbs.dtu.dk /services/TMHMM/) and r*RNA* transcripts with RNAmmer v. 1.2 (Lagesen et al., 2007).

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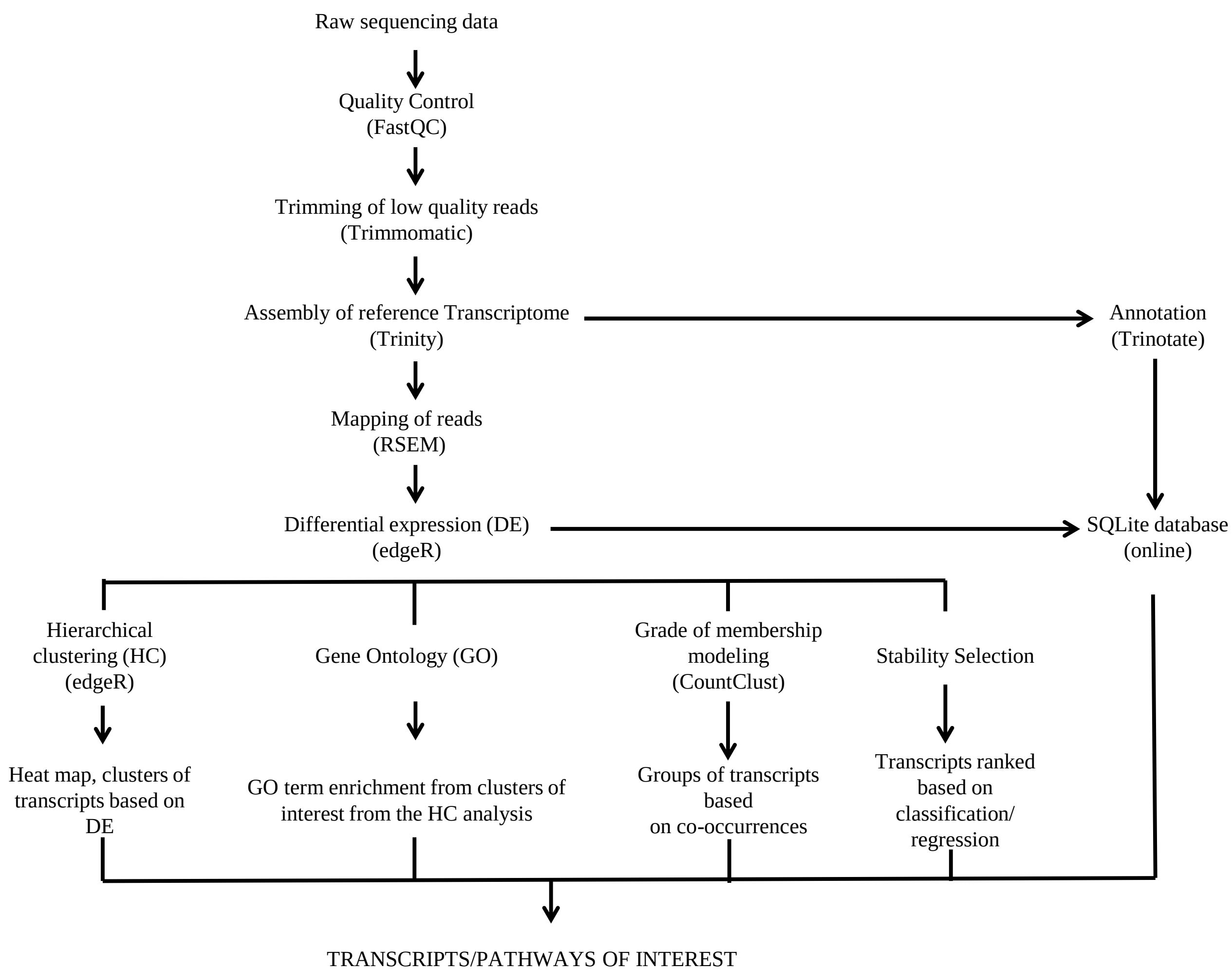
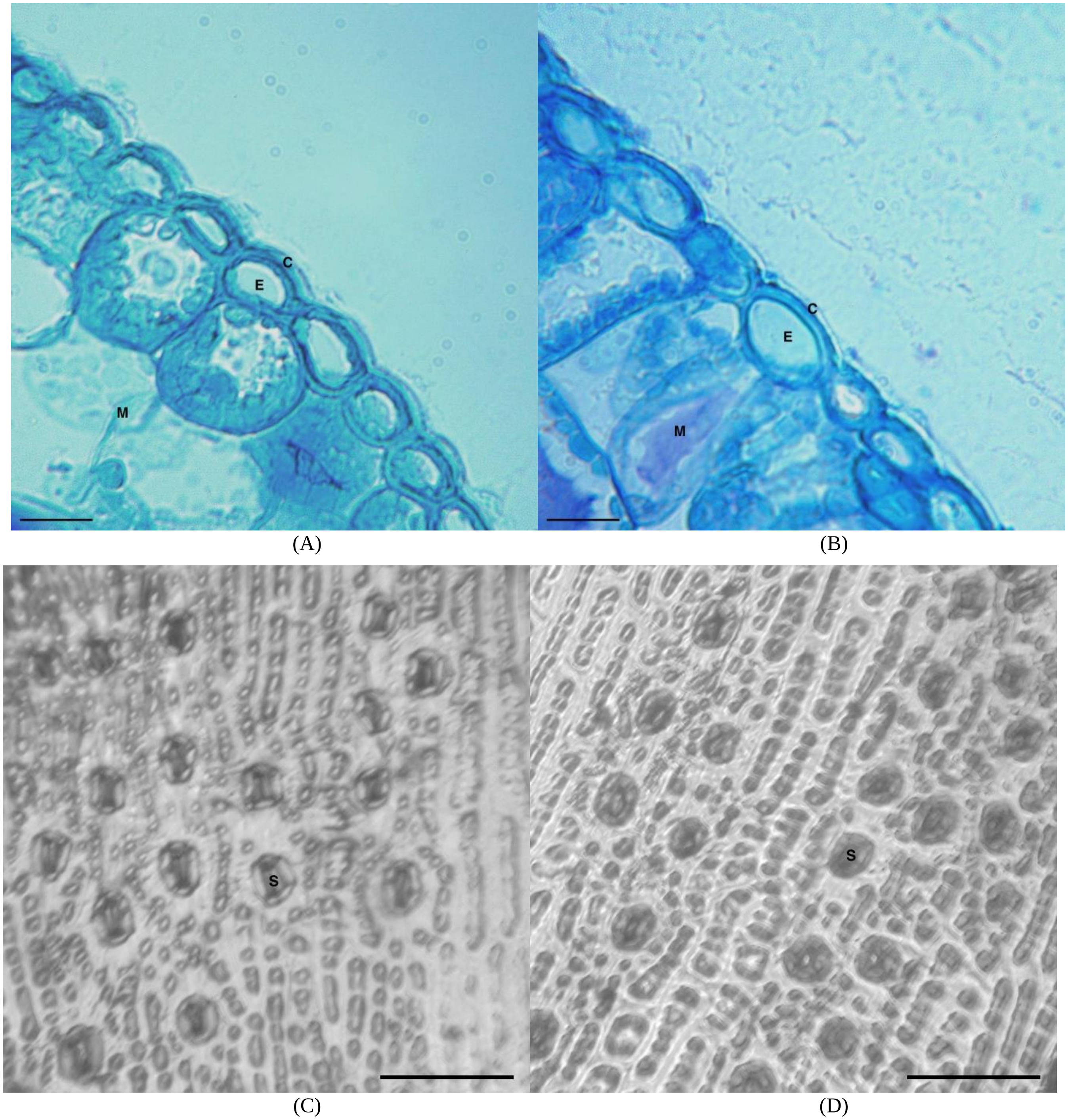
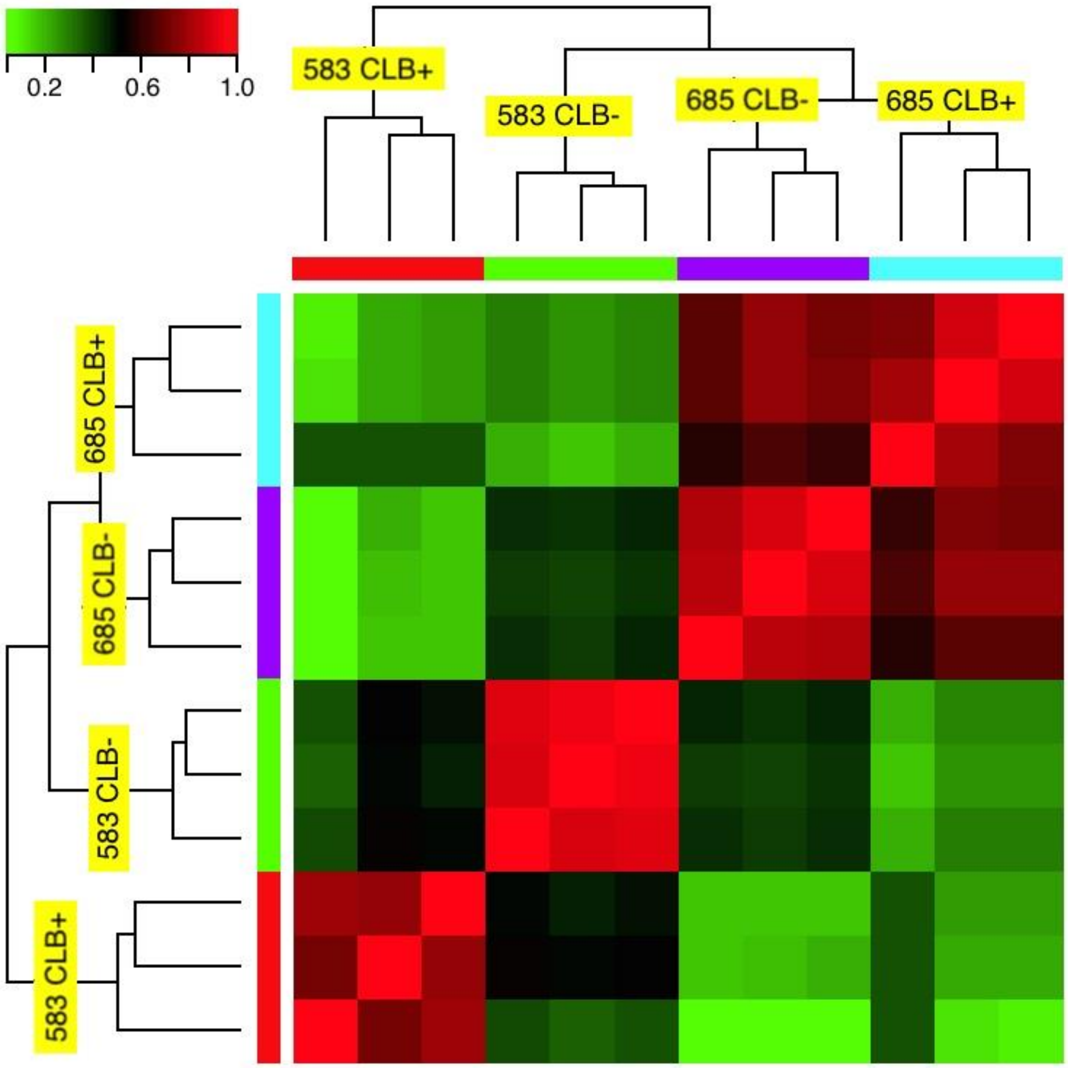


Supplementary Figure 1. Pipeline used for processing and analyzing the RNA-Seq data of the two western redcedar families in the second part of this study (see methods for details).



Supplementary Figure 2. Cuticle and stomatal density of seedlings of *Thuja plicata* full-sib families with different resistance to *Didymascella thujina*. (A) Resistant family 399 with a cuticle of 2.33 µm. (B) Susceptible family 528 with a cuticle of 2.24 µm. (C) Resistant family 685 with stomatal density of 105 stomata·mm^{-2}. (D) Susceptible family 525 with stomatal density of 118 stomata·mm^{-2}. C = cuticle, E = epidermal cell, M = mesophyll, S = stoma. Scale bars: 20 µm in a) and b), and 100 µm in c) and d).



Supplementary Figure 3. Correlation heat map of the expression profiles of all samples used in the gene expression analyses of two western redcedar families in the second part of this study. Samples were grouped using hierarchical clustering with Euclidean distance. Correlation values were colour coded according to the top left bar (green = low correlation, red = high correlation).

Supplementary Table 1. Experimental design of the study carried out in the second part of this investigation.

Item	Details
Families and conditions	·2 Families. ·2 Treatments (CLB$^+$, CLB$^-$). ·3 Replicates per family × treatment.
Seedlings' age	·Older than a year when infected. ·Close to 2 years old when sampled (symptoms had developed in the exposed).
Inoculation technique	·Exposure to *Didymascella thujina* in a *Thuja plicata* progeny trial site in Jordan River (British Columbia, Canada).
Infection confirmation	·Disease severity assessment of the infected foliage using colour imaging analysis. ·Scanning electron microscopy study of *D. thujina* spores. ·BLASTn search of the two *D. thujina*'s ITS2 in the assembled transcriptome.
Chemical analyses	Fifty-four variables: cellulose, lignin, fibre, terpenes, total C and N, non-structural carbohydrates, and mineral nutrients (see Supplementary Table 2).
RNA-Seq analyses	Illumina HiSeq 100bp paired-end technology. *In house* made libraries per sample, outsourced to Génome Québec.

Supplementary Table 3. Statistical significance (*p*-values) of the factors of the split-plot ANOVAs carried out on the 12 continuous histological variables studied (see also section 1.1 above and section 2.1.2 of the article).

	Thicknesses						**Cross section areas**		**Thickness ratios**		**Cross section area ratio**	**Lignified cell walls**
	Cuticle	**Epidermis**	**Leaf**	**Leaf width**	**Palisade mesophyll**	**Whole mesophyll**	**Leaf**	**Whole mesophyll**	**Whole mesophyll to leaf**	**Spongy to whole mesophyll**	**Whole mesophyll to leaf**	**In epidermis**
Subplots												
-Class	0.0130*	0.5552	0.1474	0.1290	0.4580	0.1441	0.1974	0.2338	0.3680	0.2350	0.9700	0.5133
-Family(Class)	0.0797	0.0011**	0.0002**	0.8240	0.0001**	0.0004**	0.0517	0.0824	0.3000	0.1390	0.8920	0.0048**
Plot												
-Crown position	0.0055**	<0.0000**	<0.0000**	0.0596	0.0018**	<0.0000**	0.0003**	0.0004**	0.0117*	<0.0000**	0.2370	0.1680
-Class × crown position	0.9421	0.7457	0.2978	0.3842	0.8091	0.3063	0.7269	0.7039	0.6068	0.3470	0.2520	0.8450
-Family(Class) × crown position	0.1200	0.0097**	0.0317*	0.3276	0.1054	0.0222*	0.8081	0.8286	0.4101	0.0430*	0.7400	0.9510

*Significant at $\alpha = 0.05$

** Significant at $\alpha = 0.01$

Supplementary Table 4. Statistics of the assembled transcriptome (calculated in PRINSEQ v. 0.20.1).

Sample size (*N*)*	12
Number of sequences	173,919
Total number of bases	134,254,425
Mean sequence length (bp)	772
Minimum sequence length (bp)	224
Maximum sequence length (bp)	16,325
Mode sequence length (bp[# seqs.])	224 (1,090)
N50 contig size (bp)	1,315

*Number of transcriptomic samples used to build the assembly.

Supplementary Table 5. Overall alignment rates of the filtered reads of all libraries used in the gene expression analyses of the second part of this study.

Family	CLB resistance class	Seedling ID	Condition	Overall alignment rate (%)
583	Susceptible	583-23	Uninfected	96.89
583	Susceptible	583-27	Uninfected	96.81
583	Susceptible	583-33	Uninfected	97.04
685	Resistant	685-24	Uninfected	96.05
685	Resistant	685-27	Uninfected	97.00
685	Resistant	685-34	Uninfected	96.80
583	Susceptible	583-2	Infected	95.93
583	Susceptible	583-12	Infected	97.19
583	Susceptible	583-17	Infected	96.00
685	Resistant	685-1	Infected	97.29
685	Resistant	685-4	Infected	97.02
685	Resistant	685-18	Infected	97.27

Supplementary Table 7. Top 50 predictors (transcripts) of the family categories (583 and 685), organized by expression cluster (see also Fig. 4 and Table 3) and decreasing stability selection score. All annotations are based on BLASTX searches done on the Swiss-Prot database (see methods for details).

Transcript	Stability selection score	Cluster in Fig. 4	E-value	Organism	Annotation	Biological process	Cellular component
TR53913|c1_g1_i4	0.0440	1	$5x10^{-74}$	*Arabidopsis* sp.	UBP1-associated protein 2A	Nucleic acid binding	
TR59400|c2_g5_i2	0.0350	1	$2x10^{-64}$	*Physcomitrella* sp.	Predicted protein		
TR57589|c7_g6_i2	0.0350	1	$3x10^{-9}$	*Picea* sp.	Putative uncharacterized protein		
TR58902|c0_g1_i2	0.0310	1	0.00	*Amborella* sp.	Uncharacterized protein	m*RNA* processing (*RNA* splicing)	
TR57804|c2_g1_i2	0.0310	1	$2x10^{-135}$	*Theobroma* sp.	Small G protein family protein / RhoGAP family protein isoform 1	Signal transduction	Intracellular
TR22003|c0_g3_i1	0.0300	1	0.00	*Oryza* sp.	Probable potassium transporter 2	Potassium ion transmembrane transporter activity	Integral component of membrane
TR58437|c7_g1_i3	0.0290	1	$7x10^{-121}$	*Arabidopsis* sp.	Zinc finger CCCH domain-containing protein 37		
TR47725|c0_g1_i1	0.0260	1	0.00	*Selaginella* sp.	Putative uncharacterized protein		
TR6396|c0_g1_i1	0.0250	1	$7x10^{-53}$	*Picea* sp.	Putative uncharacterized protein		
TR58437|c7_g1_i6	0.0240	1	$3x10^{-121}$	*Arabidopsis* sp.	Zinc finger CCCH domain-containing protein 37		
TR56381|c0_g2_i2	0.0230	1	$8x10^{-70}$	*Beta* sp.	Putative uncharacterized protein		
TR41770|c0_g1_i2	0.0230	1	$6x10^{-97}$	*Theobroma* sp.	Soul heme-binding family protein isoform 1		
TR58144|c0_g2_i5	0.0220	1	$2x10^{-31}$	*Arabidopsis* sp.	Probable leucine-rich repeat receptor-like protein kinase At1g35710		
TR59944|c4_g1_i3	0.0210	1	$3x10^{-24}$	*Pinus* sp.	Gibberellin 2-oxidase12		
TR51798|c1_g3_i1	0.0210	1	$5x10^{-55}$	*Picea* sp.	Putative uncharacterized protein		
TR48323|c0_g1_i1	0.0210	1	0.00	*Picea* sp.	Putative uncharacterized protein	Protein folding	
TR8434|c0_g2_i1	0.0200	1	$3x10^{-15}$	*Picea* sp.	Putative uncharacterized protein		
TR58308|c0_g1_i4	0.0200	1	$2x10^{-77}$	*Amborella* sp.	Uncharacterized protein		
TR57210|c8_g1_i8	0.0200	1	$3x10^{-37}$	*Fragaria* sp.	2-methylene-furan-3-one reductase		
TR45175|c0_g2_i1	0.0200	1	$1x10^{-84}$	*Picea* sp.	Putative uncharacterized protein		Cysteamine dioxygenase activity
TR20781|c1_g2_i1	0.0370	4	0.00	*Zea* sp.	Eukaryotic translation initiation factor 2 gamma subunit	Translation initiation factor activity	
TR3907|c0_g2_i1	0.0360	4	$1x10^{-55}$	*Arabidopsis* sp.	Protein ENHANCED DISEASE RESISTANCE 2		
TR52836|c0_g1_i6	0.0330	4	0.00	*Glycine* sp.	Uncharacterized protein	Regulation of signal transduction	
TR59333|c0_g1_i3	0.0290	4	$3x10^{-86}$	*Arabidopsis* sp.	Tubulin-folding cofactor C	Microtubule cytoskeleton organization	Cytoplasm
TR58279|c6_g2_i1	0.0290	4	$1x10^{-158}$	*Amborella* sp.	Uncharacterized protein	t*RNA* processing	
TR47725|c0_g1_i2	0.0280	4	0.00	*Selaginella* sp.	Putative uncharacterized protein		
TR58578|c0_g1_i9	0.0270	4	$6x10^{-77}$	*Arabidopsis* sp.	Peptidyl-tRNA hydrolase	mRNA processing (RNA splicing)	Mitochondrion
TR59409|c0_g1_i2	0.0260	4	NA	No hits	No hits		
TR58902|c0_g1_i1	0.0260	4	0.00	*Amborella* sp.	Uncharacterized protein	mRNA processing (RNA splicing)	Spliceosomal complex
TR58836|c0_g1_i2	0.0260	4	$1x10^{-101}$	*Glycine* sp.	Uncharacterized protein		
TR55562|c0_g1_i1	0.0260	4	0.00	*Oryza* sp.	ATP-dependent zinc metalloprotease FTSH 7		Chloroplast (thylakoid membrane)

TR52074|c0_g1_i1	0.0240	4	$2x10^{-19}$	*Capsella* sp.	Uncharacterized protein	Protein binding	
TR59363|c1_g3_i2	0.0230	4	NA	No hits	No hits		
TR48323|c0_g1_i2	0.0230	4	0.00	*Picea* sp.	Putative uncharacterized protein	Protein folding	
TR59228|c6_g3_i1	0.0220	4	$3x10^{-6}$	*Arabidopsis* sp.	Probable tyrosine-protein phosphatase At1g05000		
TR31863|c0_g1_i1	0.0220	4	NA	No hits	No hits		
TR52448|c0_g2_i1	0.0210	4	$2x10^{-61}$	*Selaginella* sp.	Putative uncharacterized protein	Spindle assembly	Centrosome
TR59490|c6_g2_i1	0.0200	4	NA	No hits	No hits		
TR54163|c0_g1_i1	0.0200	4	$8x10^{-98}$	*Oryza* sp.	Dephospho-CoA kinase	Coenzyme A biosynthetic process	Chloroplast, peroxisome, vacuole
TR52106|c0_g1_i2	0.0200	4	$1x10^{-93}$	*Arabidopsis* sp.	PsbP domain-containing protein 4	Photosynthesis	Photosystem II oxygen evolving complex
TR51510|c0_g1_i4	0.0200	4	$2x10^{-177}$	*Arabidopsis* sp.	Pentatricopeptide repeat-containing protein At3g62470		Mitochondrion
TR55998|c0_g1_i2	0.0190	4	$2x10^{-163}$	*Picea* sp.	Alpha-amylase	Carbohydrate metabolic process	
TR22350|c0_g1_i1	0.0190	4	$2x10^{-82}$	*Picea* sp.	Putative uncharacterized protein		
TR55562|c0_g1_i2	0.0380	6	0.00	*Oryza* sp.	ATP-dependent zinc metalloprotease FTSH 7		Chloroplast (thylakoid membrane)
TR5811|c0_g1_i1	0.0280	6	$1x10^{-139}$	*Oryza* sp.	Secretory carrier-associated membrane protein 1	Protein transport	Integral component of membrane
TR55613|c7_g3_i3	0.0270	6	$2x10^{-94}$	*Papaver* sp.	(R,S)-reticuline 7-O-methyltransferase		
TR20053|c0_g1_i1	0.0250	6	$3x10^{-32}$	*Zea* sp.	Uncharacterized protein	Response to stress	
TR33366|c0_g2_i1	0.0230	6	$4x10^{-8}$	*Picea* sp.	Putative uncharacterized protein		
TR27783|c0_g2_i1	0.0210	6	0.00	*Arabidopsis* sp.	Probable voltage-gated potassium channel subunit beta;	Potassium ion transport	Plasma membrane, plasmodesma
TR44368|c0_g2_i1	0.0260	7	$1x10^{-48}$	*Arabidopsis* sp.	Ycf20-like protein	Nonphotochemical quenching	Chloroplast

Supplementary Table 8. Representative sequences of the top transcript group per seedling from the GoM analysis. The transcripts within each group were selected based on their decreasing θ values. Disease severity refers to the percentage of foliar area that developed *D. thujina* symptoms. All annotations are based on BLASTX searches done on the Swiss-Prot database (see methods for details).

Treatment	Seedling ID	Disease severity[1]	Transcript group	θ within transcript group	Transcript	Cluster in Fig. 4	E-value	Organism	Annotation	Biological process	Cellular component
CLB$^-$	685-24	0.0000	3	0.046603901	*TR54916|c0_g2_i1*	5	NA	No hits	No hits		
				0.042521569	*TR43930|c1_g1_i1*	5	$3x10^{-42}$	*Populus* sp.	Bark storage protein A	Nucleoside metabolic process	
				0.041166356	*TR54665|c5_g1_i2*	3	0.00	*Lycopersicon* sp.	Linoleate 9S-lipoxygenase A	Oxylipin biosynthetic process	Cytoplasm
				0.024264453	*TR33201|c0_g2_i1*	5	$1x10^{-89}$	*Sedum* sp.	Peroxiredoxin Q	Antioxidant activity	Chloroplast
				0.021249167	*TR57187|c3_g1_i3*	5	$3x10^{-168}$	*Ajuga* sp.	Galactinol synthase 1	Carbohydrate storage	Cytoplasm
	685-27	0.0000	4	0.040455998	*TR54916|c0_g2_i1*	5	NA	No hits	No hits		
				0.039085152	*TR57187|c3_g1_i1*	5	$1x10^{-165}$	*Ajuga* sp.	Galactinol synthase 1	Carbohydrate storage	Cytoplasm
				0.03230091	*TR33201|c0_g2_i1*	5	$1x10^{-90}$	*Sedum* sp.	Peroxiredoxin Q	Antioxidant activity	Chloroplast
				0.032146653	*TR43930|c1_g1_i1*	5	$3x10^{-42}$	*Populus* sp.	Bark storage protein A	Nucleoside metabolic process	
				0.027273501	*TR11549|c0_g1_i1*	5	$1x10^{-56}$	*Pseudotsuga* sp.	Lea protein	Response to osmotic stress	
	685-34	0.0000	10	0.035844732	*TR57187|c3_g1_i1*	5	$1x10^{-165}$	*Ajuga* sp.	Galactinol synthase 1	Carbohydrate storage	Cytoplasm
				0.031706454	*TR59000|c3_g2_i1*	1	$3x10^{-126}$	*Arabidopsis* sp.	Cysteine-rich receptor-like protein kinase 2		
				0.03165305	*TR54916|c0_g2_i1*	5	NA	No hits	No hits		
				0.029320371	*TR56218|c1_g1_i1*	6	$2x10^{-163}$	*Triticum* sp.	RuBisCO large subunit-binding protein subunit alpha	Photosynthesis	Chloroplast
				0.028221122	*TR33201|c0_g2_i1*	5	$1x10^{-94}$	*Sedum* sp.	Peroxiredoxin Q	Antioxidant activity	Chloroplast
	583-23	0.0000	5	0.063724072	*TR54916|c0_g2_i1*	5	NA	No hits	No hits		
				0.047493443	*TR33201|c0_g2_i1*	5	$1x10^{-91}$	*Sedum* sp.	Peroxiredoxin Q	Antioxidant activity	Chloroplast
				0.044820366	*TR59130|c4_g1_i2*	9	$2x10^{-60}$	*Juniperus* sp.	Chalcone synthase	Flavonoid biosynthetic process	
				0.041464839	*TR57187|c3_g1_i1*	5	$1x10^{-165}$	*Ajuga* sp.	Galactinol synthase 1	Carbohydrate storage	Cytoplasm
				0.039524442	*TR59130|c4_g2_i3*	4	0.00	*Pinus* sp.	Chalcone synthase	Flavonoid biosynthetic process	
	583-33	0.0000	6	0.081216729	*TR26208|c0_g1_i1*	9	0.00	*Pinus* sp.	Phenylalanine ammonia-lyase	Phenyl propanoid pathway	Cytoplasm
				0.043380799	*TR54916|c0_g2_i1*	5	NA	No hits	No hits		
				0.042004155	*TR59130|c4_g2_i3*	4	0.00	*Pinus* sp.	Chalcone synthase	Flavonoid biosynthetic process	
				0.040125028	*TR59130|c4_g1_i2*	9	$2x10^{-60}$	*Juniperus* sp.	Chalcone synthase	Flavonoid biosynthetic process	
				0.03712881	*TR57187|c3_g1_i1*	5	$1x10^{-165}$	*Ajuga* sp.	Galactinol synthase 1	Carbohydrate storage	Cytoplasm
	583-27	0.0000	7	0.054058741	*TR54916|c0_g2_i1*	5	NA	No hits	No hits		
				0.046281105	*TR33201|c0_g2_i1*	5	$1x10^{-93}$	*Sedum* sp.	Peroxiredoxin Q	Antioxidant activity	Chloroplast
				0.044402383	*TR57187|c3_g1_i1*	5	$1x10^{-165}$	*Ajuga* sp.	Galactinol synthase 1	Carbohydrate storage	Cytoplasm
				0.038447621	*TR26208|c0_g1_i1*	9	0.00	*Pinus* sp.	Phenylalanine ammonia-lyase	Phenyl propanoid pathway	Cytoplasm
				0.030467396	*TR59236|c4_g1_i4*	5	NA	No hits	No hits		

CLB^{+}	685-18	2.0463	1	0.081751137	*TR65408|c0_g1_i1*	2	$9x10^{-43}$	*Arabidopsis* sp.	Early light-induced protein 1		Chloroplast
				0.064815572	*TR9335|c0_g1_i1*	2	$1x10^{-45}$	*Arabidopsis* sp.	Early light-induced protein 1		Chloroplast
				0.042225173	*TR19072|c0_g1_i1*	2	$9x10^{-55}$	Oryza sp.	Tricin synthase 1	Biosynthesis of secondary metabolites	Nucleus
				0.035711716	*TR59000|c3_g2_i1*	1	$3x10^{-123}$	*Arabidopsis* sp.	Cysteine-rich receptor-like protein kinase 2		
				0.028796717	*TR9374|c0_g1_i1*	2	$3x10^{-82}$	*Arabidopsis* sp.	Cinnamoyl-CoA reductase 1	Phenylpropanoid pathway	
	685-4	0.6789	8	0.067591526	*TR65408|c0_g1_i1*	2	$9x10^{-43}$	*Arabidopsis* sp.	Early light-induced protein 1		Chloroplast
				0.055552706	*TR9335|c0_g1_i1*	2	$1x10^{-45}$	*Arabidopsis* sp.	Early light-induced protein 1		Chloroplast
				0.041974572	*TR19072|c0_g1_i1*	2	$9x10^{-55}$	*Oryza* sp.	Tricin synthase 1	Biosynthesis of secondary metabolites	Nucleus
				0.032156825	*TR9374|c0_g1_i1*	2	$3x10^{-84}$	*Arabidopsis* sp.	Cinnamoyl-CoA reductase 1	Phenylpropanoid pathway	
				0.014753053	*TR59000|c3_g2_i1*	1	$3x10^{-125}$	*Arabidopsis* sp.	Cysteine-rich receptor-like protein kinase 2		
	685-1	1.0326	12	0.072317276	*TR65408|c0_g1_i1*	2	$9x10^{-43}$	*Arabidopsis* sp.	Early light-induced protein 1		Chloroplast
				0.044026819	*TR9335|c0_g1_i1*	2	$1x10^{-45}$	*Arabidopsis* sp.	Early light-induced protein 1		Chloroplast
				0.040247343	*TR19072|c0_g1_i1*	2	$9x10^{-55}$	*Oryza* sp.	Tricin synthase 1	Biosynthesis of secondary metabolites	Nucleus
				0.029319831	*TR9374|c0_g1_i1*	2	$3x10^{-86}$	*Arabidopsis* sp.	Cinnamoyl-CoA reductase 1	Phenylpropanoid pathway	
				0.017502319	*TR56218|c1_g1_i1*	6	$2x10^{-163}$	*Triticum* sp.	RuBisCO large subunit-binding protein subunit alpha	Photosynthesis	Chloroplast
	583-17	3.5383	9	0.051165284	*TR49492|c0_g1_i1*	8	$3x10^{-35}$	*Neurospora* sp.	Uncharacterized protein		
				0.047667562	*TR65408|c0_g1_i1*	2	$9x10^{-43}$	*Arabidopsis* sp.	Early light-induced protein 1		Chloroplast
				0.036443332	*TR19072|c0_g1_i1*	2	$9x10^{-55}$	*Oryza* sp.	Tricin synthase 1	Biosynthesis of secondary metabolites	Nucleus
				0.027368645	*TR9335|c0_g1_i1*	2	$1x10^{-45}$	*Arabidopsis* sp.	Early light-induced protein 1		Chloroplast
				0.025274788	*TR59130|c4_g1_i2*	9	$2x10^{-60}$	*Juniperus* sp.	Chalcone synthase	Flavonoid biosynthetic process	
	583-12	4.3511	11	0.052234732	*TR65408|c0_g1_i1*	2	$9x10^{-43}$	*Arabidopsis* sp.	Early light-induced protein 1		Chloroplast
				0.050508529	*TR26208|c0_g1_i1*	9	0.00	*Pinus* sp.	Phenylalanine ammonia-lyase	Phenyl propanoid pathway	Cytoplasm
				0.045791771	*TR53697|c0_g2_i1*	7	$2x10^{-129}$	*Cryptomeria* sp.	Thaumatin-like protein		
				0.041779946	*TR59130|c4_g2_i3*	4	0.00	*Pinus* sp.	Chalcone synthase	Flavonoid biosynthetic process	
				0.039400874	*TR58068|c0_g1_i1*	8	$4x10^{-127}$	*Cryptomeria* sp.	Pollen allergen CJP38	Hydrolase activity, hydrolyzing O-glycosyl compounds	
	583-2	3.4456	13	0.125856284	*TR54916|c0_g2_i1*	5	NA	No hits	No hits		
				0.059133626	*TR59043|c3_g1_i2*	8	$2x10^{-124}$	*Juniperus* sp.	Pathogenesis-related protein	Defense response	
				0.049384925	*TR59043|c3_g1_i1*	8	$9x10^{-138}$	*Juniperus* sp.	Pathogenesis-related protein	Defense response	
				0.030879161	*TR43930|c1_g1_i1*	5	$3x10^{-42}$	*Populus* sp.	Bark storage protein A	Nucleoside metabolic process	
				0.028219645	*TR11549|c0_g1_i1*	5	$1x10^{-56}$	*Pseudotsuga* sp.	Lea protein	Response to osmotic stress	